

Assignment: Phylogenetic Analysis of Dengue Virus Genomes Using Galaxy

Module

Bioinformatics

Assignment Title

Construction and Interpretation of a Dengue Virus Phylogenetic Tree Using Multiple Sequence Alignment and Maximum Likelihood Analysis

Background

Phylogenetic analysis is a fundamental bioinformatics technique used to study evolutionary relationships among organisms and viruses. During outbreaks, phylogenetic trees help researchers understand viral evolution, transmission patterns, emergence of new strains, and geographical spread.

In this assignment, students will obtain Dengue virus genome sequences from the NCBI Virus database, perform multiple sequence alignment using MAFFT, and construct a phylogenetic tree using IQ-TREE within the Galaxy platform.

Learning Outcomes

Upon successful completion of this assignment, students will be able to:

1. Retrieve biological sequence data from public repositories.
2. Apply multiple sequence alignment techniques to viral genomes.
3. Construct phylogenetic trees using Maximum Likelihood methods.
4. Interpret evolutionary relationships among viral strains.
5. Utilize Galaxy workflows for reproducible bioinformatics analyses.

Assignment Tasks

Task 1: Sequence Retrieval from NCBI Virus Database

Access the NCBI Virus database and retrieve Dengue virus genome sequences using the following criteria:

Search Criteria

- Virus: Dengue virus
- Sequence Type: Complete Genome

- Host: Human
- Isolation Source: Blood
- Geographic Region: Asia

Download a minimum of 20 complete genome sequences in FASTA format.

Record:

- Number of sequences retrieved
- Countries represented
- Collection dates (if available)

Include screenshots of the search filters used.

Task 2: Data Preparation

Review the downloaded FASTA file and ensure:

- All sequences are complete genomes
- Sequences are in FASTA format
- Sequence identifiers are preserved

Provide:

- Number of sequences analyzed
- Average sequence length
- Example FASTA entry

Task 3: Multiple Sequence Alignment Using MAFFT

Using Galaxy:

1. Upload the FASTA file.
2. Run MAFFT.
3. Use default parameters unless otherwise justified.
4. Download the aligned FASTA output.

Include:

- Screenshot of MAFFT workflow execution

- Alignment statistics
- Description of alignment purpose

Task 4: Phylogenetic Tree Construction Using IQ-TREE

Using Galaxy:

1. Input the MAFFT alignment into IQ-TREE.
2. Use Maximum Likelihood tree construction.
3. Perform bootstrap analysis with 1000 replicates.
4. Generate the phylogenetic tree.

Record:

- Evolutionary model selected by IQ-TREE
- Bootstrap settings
- Tree file generated

Include screenshots.

Task 5: Tree Visualization and Interpretation

Visualize the resulting phylogenetic tree.

Analyze:

1. Number of major clusters observed.
2. Whether strains from the same country cluster together.
3. Presence of distinct evolutionary lineages.
4. Branches with strong bootstrap support (>70%).
5. Possible explanations for observed clustering patterns.

Task 6: Discussion

Discuss the following:

Question 1

Why is sequence alignment required before phylogenetic analysis?

Question 2

What is the purpose of bootstrap analysis?

Question 3

How does the Maximum Likelihood approach differ from distance-based methods?

Question 4

What biological conclusions can be drawn from the generated tree?

Question 5

What are the limitations of this analysis?

Report Structure

1. Introduction
2. Materials and Methods
3. Results
4. Discussion
5. Conclusion
6. References

Expected report length: 1500–2500 words.

Submission Requirements

Submit:

- PDF report
- Original FASTA file
- MAFFT alignment output
- IQ-TREE output files
- Galaxy workflow export

Marking Scheme (100 Marks)

Component	Marks
Data Retrieval and Documentation	15
Data Preparation	10
MAFFT Alignment	15
IQ-TREE Analysis	20
Tree Interpretation	20
Discussion and Biological Understanding	10
Report Presentation and References	10
Total	100

Bonus Challenge (+10 Marks)

Compare phylogenetic trees generated using:

- 100 bootstrap replicates
- 1000 bootstrap replicates

Discuss how bootstrap values influence confidence in evolutionary relationships.

Expected Deliverable

A reproducible Galaxy workflow that downloads Dengue virus genome sequences, performs multiple sequence alignment using MAFFT, constructs a phylogenetic tree using IQ-TREE, and provides a biological interpretation of the observed evolutionary relationships among Asian human-derived Dengue virus isolates.